



Technology Data Sheet (TDS)

TINMORRY PP-CF

Version 1.0

• Basic Introduction

TINMORRY PP-CF is a carbon fiber (CF) reinforced PP composite material. Like generic PP, it has outstanding basic properties such as low density, high toughness, bending resistance, and chemical corrosion resistance. In addition, thanks to the addition of carbon fiber and the specially designed and optimized formula, it can endow the prints with a delicate, smooth appearance with invisible layer lines and excellent layer adhesion, making the prints both beautiful and durable. Therefore, TINMORRY PP-CF is suitable for printing lightweight, impact-resistant, chemical-resistant engineering structural parts, mechanical parts, and delicate decorations and exhibits.

Note: 1. Please be sure to use **glue for PP** to treat the printing plate first to prevent the models from warping or falling off during printing. 2. Since the PP-CF filament contains carbon fiber, please use a nozzle with a diameter of ≥ 0.4 mm to decrease the risk of nozzle clogging; and please use hard enough nozzles such as those made of hardened steel, and do not use relatively soft nozzles such as those made of brass or stainless steel to slow down the wear of them.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	700 g $\pm 0.5\%$
Length	320 ± 5 m
Spool material	ABS ((Temperature resistance: 80 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	60 - 75 °C, 6 - 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	$\leq 40\text{ °C}$, $\leq 20\%$ RH (Put in an air-tight container and sealed with desiccant)
Storage temperature and humidity	$\leq 30\text{ °C}$, $\leq 20\%$ RH (Sealed with desiccant)
Compatible support material	Itself or support filament for PP
Compatible printer type	Enclosed-frame, open-frame
Compatible nozzle material	Hardened steel
Compatible nozzle size	0.4 (recommended), 0.6, 0.8 mm
Compatible plate type	Smooth / Textured PEI Plate or other plate
Plate surface preparation	Glue for PP
Nozzle temperature	230 - 280 °C
Bed temperature	80 - 110 °C (depending on the printer and plate types)
Printing speed*	Max: 150 mm/s; Recommended: $\leq 130\text{ mm/s}$
Chamber temperature	$< 45\text{ °C}$ (when the temperature is higher than 40 °C, opening the front door and upper lid is needed to avoid extruder clogging)

*When printed with Bambu X1C and P1S, and the nozzle temperature is 245 and 250 °C, the max printing speed can reach 130 and 150 mm/s.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY PP-CF have been tested and are shown in the tables below. The annealing parameters of the specimens is as followed: 60 °C for 6 h → 70 °C for 8 h, and then cool down to room temperature slowly.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	0.91 g/cm ³
Saturated water absorption rate	25 °C, 55% RH	~ 0.05%
Melt index	230 °C, 2.16 kg	8.5 ± 0.6 g/ 10 min
Glass transition temperature	DSC, 10 °C/min	- 1 °C
Crystallization temperature	DSC, 10 °C/min	116 °C
Melting temperature	DSC, 10 °C/min	152 °C
Vicar softening temperature	ISO 306, GB/T 1633	68 °C Annealed: 77 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	46 °C Annealed: 49 °C

Heat deflection temperature 2	ISO 75, 0.45 MPa	60 °C Annealed: 70 °C
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• **Mechanical Properties**

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	26 ± 2 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	23 ± 2 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	1356 ± 126 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	972 ± 141 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	158% ~ 182%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	52% ~ 65%
Bending strength (XY)	ISO 178, GB/T 9341	24 ± 2 MPa
Bending strength (Z)	ISO 178, GB/T 9341	22 ± 2 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	1238 ± 115 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	826 ± 75 MPa
Impact strength (XY)	ISO 179, GB/T 1043	61.3 ± 2.5 kJ/m ² ; Annealed: 100.6 ± 6.1 kJ/m ²
Impact strength (Z)	ISO 179, GB/T 1043	16.4 ± 1.4 kJ/m ²

• **Chemical and Other Properties**

Subjects	Data
Component	PP, carbon fiber
Skin irritation	May cause allergy in some people, and wearing gloves is suggested
Resistance to water	Yes
Resistance to oil and grease	Resistant to most kinds of daily oil and grease at room temperature
Resistance to organic solvent	Resistant to most non-polar and weakly polar organic solvents at room temperature, but not resistant to some organic solvents at high temperature
Resistance to weak acid	Yes
Resistance to strong acid	Not resistant to strong oxidizing acids such as concentrated sulfuric acid and concentrated nitric acid
Resistance to weak alkali	Yes
Resistance to strong alkali	Not resistant when for a long time
Flammability	Yes
Odor during printing	Odorless
Odor during combustion	Similar to the smell of burning candles and stronger

• **Specimen Info**

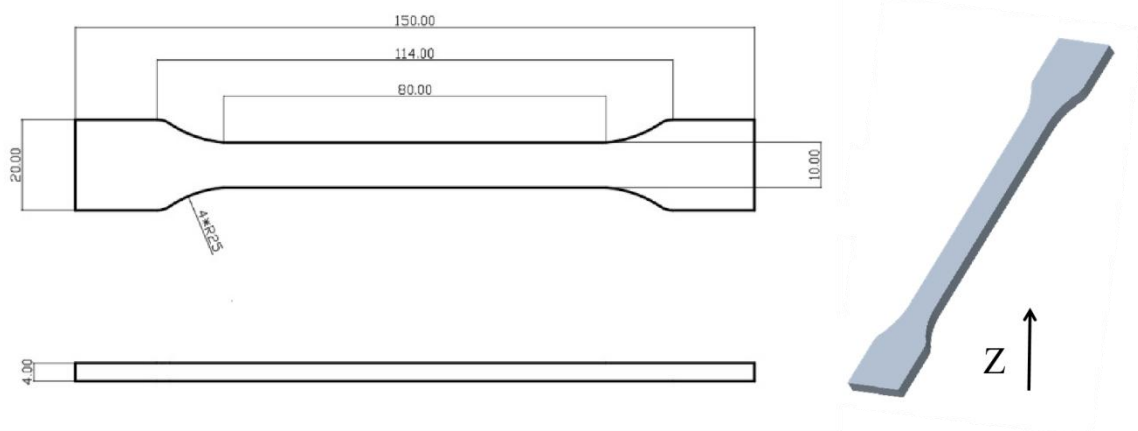
Subjects	Data
Size	Shown at the following pictures

Nozzle temperature	245 °C
Bed temperature	60 °C
Printing speed	XY: 100 mm/s; Z: 80 mm/s
Infill density	100%
Infill pattern	Concentric

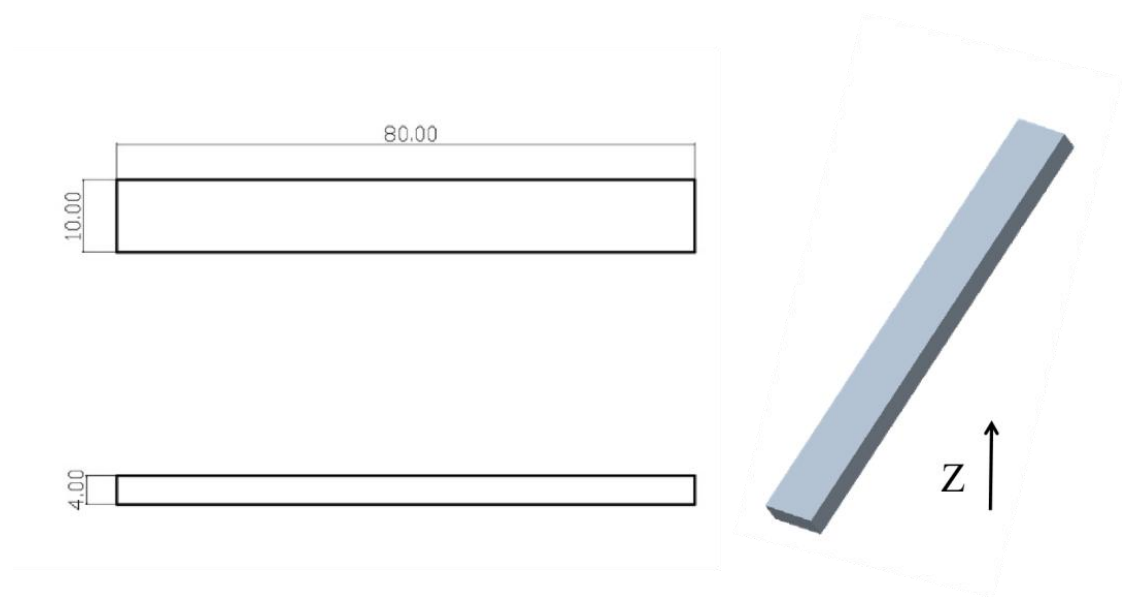
Statements:

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, the impact strength of TINMORRY PP-CF prints can be slightly increased and other mechanical properties hardly change, while some prints may undergo shrinkage or deformation, so please be aware of this. The recommended annealing process is: first at 60 °C for 6 hours, then slowly heat up to about 70 °C (not recommended to exceed 75 °C) for about 8 hours, and finally slowly cool down in the oven. Prints of different sizes, structures, and shapes can have different annealing performances. If the print is found to be significantly deformed during the process, please switch to a lower annealing temperature.

1. Tensile Test



2. Bending Test and Impact Test



• Disclaimer

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them, or the usage scenarios of them.